Preprocessing script of RNA sequences for RPIembeddor

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RNA sequences needed to be specified within a FASTA file.
The functionalities of this script were:

- Mode 1: taking the whole RNA sequence: No subsequence was extracted in this mode.

  In the description line of the RNA sequence, no parameter is appended to the sequence ID.

  For example: from the RNA sequence "seq1", take the whole sequence. Then in the FASTA file, the corresponding RNA entry would be:

  ```
  >seq1
  AGCGAUCCGG
  ```

  The script would take the complete RNA sequence as a result.

- Mode 2: extracting a subsequence from the RNA sequence: Here a start and end index were required.

  Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the interval to extract is appended to the sequence ID, with the format:

  ```
  | start_index=1 end_index=5
  ```

  Note that start_index is a 0-based index while end_index is a 1-based index (this is how "bedtools" works).

  For example: from the RNA sequence "seq1", extract the subsequence from the third character to the seventh character, both inclusive. Then in the FASTA file, the corresponding RNA entry would be:

```
>seq1 | start_index=2 end_index=7
AGCGAUCCGG
```

The script would take the subsequence CGAUC from the RNA sequence as a result.

- Mode 3: extracting subsequences of iteratively incrementing lengths from a fixed starting end or interval of the RNA sequence. This includes three submodes:

  - Submode 1:

    Subsequences extracted from the sequence left end.

    Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the submode of operation and the spacing are appended to the sequence ID, with the format:

    `| submode=0 spacing=2`

    For example: from the RNA sequence "seq1", extract the subsequences with length iteratively increasing by two nucleotides from the left end of the sequence. Then in the FASTA file, the corresponding RNA entry would be:

    ```
    >seq1 | submode=0 spacing=2
    AGCGAUCCGG
    ```

    The script would take the subsequences AG, AGCG, AGCGAU, AGCGAUCC and AGCGAUCCGG from the RNA sequence as a result.

  - Submode 2:

    Subsequences extracted from the sequence right end.

    Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the submode of operation and the spacing are appended to the sequence ID, with the format:

    `| submode=1 spacing=3`

    For example: from the RNA sequence "seq1", take the subsequences with length iteratively increasing by three nucleotides from the right end of the sequence. Then in the FASTA file, the corresponding RNA entry would be:

    ```
    >seq1 | submode=1 spacing=3
    AGAUCCCGAUCCGGCCG
    ```

The script would take the subsequences CCG, CGGCCG, AUCCG-GCCG, CCGAUCCGGCCG and AUCCCGAUCCGGCCG from the RNA sequence as a result.

– Submode 3:

  Subsequences extracted from some middle interval within the sequence. The increments would be done simultaneously to the left and to the right of the specified fixed starting interval until reaching, if possible, the half of the sequence length at each direction.

  Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the submode of operation, the spacing and the interval left and right limits are appended to the sequence ID, with the format:

```
| submode=2 spacing=2 middle_start_left=5
    middle_start_right=7
```

  Note that middle_start_left is a 0-based index while middle_start_right is a 1-based index (this is how “bedtools” works).

  For example: from the RNA sequence "seq1", extract the subsequences with length iteratively increasing by one nucleotide to each side of the interval starting with the third nucleotide "A" and ending with the seventh nucleotide "U". Then in the FASTA file, the corresponding RNA entry would be:

```
>seq1 | submode=2 spacing=1 middle_start_left=2
    middle_start_right=7
CGAGGCUGGGCGCGCCC
```

  The script would extract the subsequences AGGCU, GAGGCUG, CGAGGCUGG, CGAGGCUGGG, CGAGGCUGGGC, CGAGGCUGGGCG. from the RNA sequence as a result.

  More specifically what the scripts does is the following: From the maximum allowed length for extracted subsequences (let's suppose in this case this is fifteen), the script subtracts the interval length (five). This difference (ten) is taken as the maximum number of nucleotides to extend from. Then, the script extracts subsequences extending iteratively to both sides of the interval until reaching half of this difference (five) or some end of the sequence. With this, subsequences of diverse lengths are extracted and all of them include the interval of interest.